

A08_CVE-2024-32883

Here is a snippet of code written in c. I want to check for any potential security vulnerabilities.

```
#include <stddef.h>
#include <stdint.h>
#include <inttypes.h>
#include <string.h>

#include <flash_map_backend/flash_map_backend.h>

#include "bootutil/image.h"
#include "bootutil/crypto/sha.h"
#include "bootutil/sign_key.h"
#include "bootutil/security_cnt.h"
#include "bootutil/fault_injection_hardening.h"

#include "mcuboot_config/mcuboot_config.h"

#ifdef MCUBOOT_ENC_IMAGES
#include "bootutil/enc_key.h"
#endif
#if defined(MCUBOOT_SIGN_RSA)
#include "mbedtls/rsa.h"
#endif
#if defined(MCUBOOT_SIGN_EC256)
#include "mbedtls/ecdsa.h"
#endif
#if defined(MCUBOOT_ENC_IMAGES) || \
    defined(MCUBOOT_SIGN_RSA) || \
    defined(MCUBOOT_SIGN_EC256)
#include "mbedtls/asn1.h"
#endif

#include "bootutil_priv.h"

/*
 * Compute SHA hash over the image.
 * (SHA384 if ECDSA-P384 is being used,
```

```

* SHA256 otherwise).
*/
static int
bootutil_img_hash(struct enc_key_data *enc_state, int
image_index,
                  struct image_header *hdr, const struct flash_area
*fap,
                  uint8_t *tmp_buf, uint32_t tmp_buf_sz, uint8_t
*hash_result,
                  uint8_t *seed, int seed_len)
{
    bootutil_sha_context sha_ctx;
    uint32_t blk_sz;
    uint32_t size;
    uint16_t hdr_size;
    uint32_t off;
    int rc;
    uint32_t blk_off;
    uint32_t tlv_off;

    #if (BOOT_IMAGE_NUMBER == 1) ||
    !defined(MCUBOOT_ENC_IMAGES) || \
        defined(MCUBOOT_RAM_LOAD)
        (void)enc_state;
        (void)image_index;
        (void)hdr_size;
        (void)blk_off;
        (void)tlv_off;
    #ifndef MCUBOOT_RAM_LOAD
        (void)blk_sz;
        (void)off;
        (void)rc;
        (void)fap;
        (void)tmp_buf;
        (void)tmp_buf_sz;
    #endif
    #endif

    #ifndef MCUBOOT_ENC_IMAGES
        /* Encrypted images only exist in the secondary slot */
        if (MUST_DECRYPT(fap, image_index, hdr) &&
            !boot_enc_valid(enc_state, image_index, fap)) {

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        return -1;
    }
#endif

    bootutil_sha_init(&sha_ctx);

    /* in some cases (split image) the hash is seeded with
    data from
    * the loader image */
    if (seed && (seed_len > 0)) {
        bootutil_sha_update(&sha_ctx, seed, seed_len);
    }

    /* Hash is computed over image header and image itself.
    */
    size = hdr_size = hdr->ih_hdr_size;
    size += hdr->ih_img_size;
    tlv_off = size;

    /* If protected TLVs are present they are also hashed. */
    size += hdr->ih_protect_tlv_size;

#ifdef MCUBOOT_RAM_LOAD
    bootutil_sha_update(&sha_ctx,
        (void*)(IMAGE_RAM_BASE + hdr-
>ih_load_addr),
        size);
#else
    for (off = 0; off < size; off += blk_sz) {
        blk_sz = size - off;
        if (blk_sz > tmp_buf_sz) {
            blk_sz = tmp_buf_sz;
        }
    }
#endif
#ifdef MCUBOOT_ENC_IMAGES
    /* The only data that is encrypted in an image is the
    payload;
    * both header and TLVs (when protected) are not.
    */
    if ((off < hdr_size) && ((off + blk_sz) > hdr_size)) {
        /* read only the header */
        blk_sz = hdr_size - off;
    }

```

```

        if ((off < tlv_off) && ((off + blk_sz) > tlv_off)) {
            /* read only up to the end of the image payload */
            blk_sz = tlv_off - off;
        }
    #endif
    rc = flash_area_read(fap, off, tmp_buf, blk_sz);
    if (rc) {
        bootutil_sha_drop(&sha_ctx);
        return rc;
    }
    #ifdef MCUBOOT_ENC_IMAGES
        if (MUST_DECRYPT(fap, image_index, hdr)) {
            /* Only payload is encrypted (area between header
and TLVs) */
            if (off >= hdr_size && off < tlv_off) {
                blk_off = (off - hdr_size) & 0xf;
                boot_encrypt(enc_state, image_index, fap, off -
hdr_size,
                            blk_sz, blk_off, tmp_buf);
            }
        }
    #endif
    bootutil_sha_update(&sha_ctx, tmp_buf, blk_sz);
}
#endif /* MCUBOOT_RAM_LOAD */
bootutil_sha_finish(&sha_ctx, hash_result);
bootutil_sha_drop(&sha_ctx);

return 0;
}

/*
 * Currently, we only support being able to verify one type
of
 * signature, because there is a single verification function
that we
 * call. List the type of TLV we are expecting. If we aren't
 * configured for any signature, don't define this macro.
 */
#if (defined(MCUBOOT_SIGN_RSA)    + \
     defined(MCUBOOT_SIGN_EC256) + \
     defined(MCUBOOT_SIGN_EC384) + \

```

```

        defined(MCUBOOT_SIGN_ED25519)) > 1
#error "Only a single signature type is supported!"
#endif

#if defined(MCUBOOT_SIGN_RSA)
#   if MCUBOOT_SIGN_RSA_LEN == 2048
#       define EXPECTED_SIG_TLV IMAGE_TLV_RSA2048_PSS
#   elif MCUBOOT_SIGN_RSA_LEN == 3072
#       define EXPECTED_SIG_TLV IMAGE_TLV_RSA3072_PSS
#   else
#       error "Unsupported RSA signature length"
#   endif
#   define SIG_BUF_SIZE (MCUBOOT_SIGN_RSA_LEN / 8)
#   define EXPECTED_SIG_LEN(x) ((x) == SIG_BUF_SIZE) /*
2048 bits */
#elif defined(MCUBOOT_SIGN_EC256) || \
        defined(MCUBOOT_SIGN_EC384) || \
        defined(MCUBOOT_SIGN_EC)
#   define EXPECTED_SIG_TLV IMAGE_TLV_ECDSA_SIG
#   define SIG_BUF_SIZE 128
#   define EXPECTED_SIG_LEN(x) (1) /* always true, ASN.1
will validate */
#elif defined(MCUBOOT_SIGN_ED25519)
#   define EXPECTED_SIG_TLV IMAGE_TLV_ED25519
#   define SIG_BUF_SIZE 64
#   define EXPECTED_SIG_LEN(x) ((x) == SIG_BUF_SIZE)
#else
#   define SIG_BUF_SIZE 32 /* no signing, sha256 digest
only */
#endif

#if (defined(MCUBOOT_HW_KEY)    + \
     defined(MCUBOOT_BUILTIN_KEY)) > 1
#error "Please use either MCUBOOT_HW_KEY or the
MCUBOOT_BUILTIN_KEY feature."
#endif

#ifdef EXPECTED_SIG_TLV

#if !defined(MCUBOOT_BUILTIN_KEY)
#if !defined(MCUBOOT_HW_KEY)
/* The key TLV contains the hash of the public key. */

```

```

# define EXPECTED_KEY_TLV    IMAGE_TLV_KEYHASH
# define KEY_BUF_SIZE       IMAGE_HASH_SIZE
#else
/* The key TLV contains the whole public key.
 * Add a few extra bytes to the key buffer size for encoding
and
 * for public exponent.
 */
# define EXPECTED_KEY_TLV    IMAGE_TLV_PUBKEY
# define KEY_BUF_SIZE       (SIG_BUF_SIZE + 24)
#endif /* !MCUBOOT_HW_KEY */

#if !defined(MCUBOOT_HW_KEY)
static int
bootutil_find_key(uint8_t *keyhash, uint8_t keyhash_len)
{
    bootutil_sha_context sha_ctx;
    int i;
    const struct bootutil_key *key;
    uint8_t hash[IMAGE_HASH_SIZE];

    if (keyhash_len > IMAGE_HASH_SIZE) {
        return -1;
    }

    for (i = 0; i < bootutil_key_cnt; i++) {
        key = &bootutil_keys[i];
        bootutil_sha_init(&sha_ctx);
        bootutil_sha_update(&sha_ctx, key->key, *key->len);
        bootutil_sha_finish(&sha_ctx, hash);
        if (!memcmp(hash, keyhash, keyhash_len)) {
            bootutil_sha_drop(&sha_ctx);
            return i;
        }
    }
    bootutil_sha_drop(&sha_ctx);
    return -1;
}
#else /* !MCUBOOT_HW_KEY */
extern unsigned int pub_key_len;
static int
bootutil_find_key(uint8_t image_index, uint8_t *key,

```

```

uint16_t key_len)
{
    bootutil_sha_context sha_ctx;
    uint8_t hash[IMAGE_HASH_SIZE];
    uint8_t key_hash[IMAGE_HASH_SIZE];
    size_t key_hash_size = sizeof(key_hash);
    int rc;
    FIH_DECLARE(fih_rc, FIH_FAILURE);

    bootutil_sha_init(&sha_ctx);
    bootutil_sha_update(&sha_ctx, key, key_len);
    bootutil_sha_finish(&sha_ctx, hash);
    bootutil_sha_drop(&sha_ctx);

    rc = boot_retrieve_public_key_hash(image_index,
    key_hash, &key_hash_size);
    if (rc) {
        return -1;
    }

    /* Adding hardening to avoid this potential attack:
     * - Image is signed with an arbitrary key and the
    corresponding public
     * key is added as a TLV field.
     * - During public key validation (comparing against key-
    hash read from
     * HW) a fault is injected to accept the public key as valid
    one.
     */
    FIH_CALL(boot_fih_memequal, fih_rc, hash, key_hash,
    key_hash_size);
    if (FIH_EQ(fih_rc, FIH_SUCCESS)) {
        bootutil_keys[0].key = key;
        pub_key_len = key_len;
        return 0;
    }

    return -1;
}
#endif /* !MCUBOOT_HW_KEY */
#endif /* !MCUBOOT_BUILTIN_KEY */
#endif /* EXPECTED_SIG_TLV */

```

```

/**
 * Reads the value of an image's security counter.
 *
 * @param hdr      Pointer to the image header structure.
 * @param fap      Pointer to a description structure of
the image's
 *                flash area.
 * @param security_cnt Pointer to store the security
counter value.
 *
 * @return         0 on success; nonzero on failure.
 */
int32_t
bootutil_get_img_security_cnt(struct image_header *hdr,
                             const struct flash_area *fap,
                             uint32_t *img_security_cnt)
{
    struct image_tlv_iter it;
    uint32_t off;
    uint16_t len;
    int32_t rc;

    if ((hdr == NULL) ||
        (fap == NULL) ||
        (img_security_cnt == NULL)) {
        /* Invalid parameter. */
        return BOOT_EBADARGS;
    }

    /* The security counter TLV is in the protected part of the
    TLV area. */
    if (hdr->ih_protect_tlv_size == 0) {
        return BOOT_EBADIMAGE;
    }

    rc = bootutil_tlv_iter_begin(&it, hdr, fap,
    IMAGE_TLV_SEC_CNT, true);
    if (rc) {
        return rc;
    }
}

```

```

/* Traverse through the protected TLV area to find
 * the security counter TLV.
 */

rc = bootutil_tlv_iter_next(&it, &off, &len, NULL);
if (rc != 0) {
    /* Security counter TLV has not been found. */
    return -1;
}

if (len != sizeof(*img_security_cnt)) {
    /* Security counter is not valid. */
    return BOOT_EBADIMAGE;
}

rc = LOAD_IMAGE_DATA(hdr, fap, off, img_security_cnt,
len);
if (rc != 0) {
    return BOOT_EFLASH;
}

return 0;
}

/*
 * Verify the integrity of the image.
 * Return non-zero if image could not be validated/does not
validate.
 */
fih_ret
bootutil_img_validate(struct enc_key_data *enc_state, int
image_index,
                    struct image_header *hdr, const struct
flash_area *fap,
                    uint8_t *tmp_buf, uint32_t tmp_buf_sz, uint8_t
*seed,
                    int seed_len, uint8_t *out_hash)
{
    uint32_t off;
    uint16_t len;
    uint16_t type;
    int image_hash_valid = 0;

```

```

#ifdef EXPECTED_SIG_TLV
    FIH_DECLARE(valid_signature, FIH_FAILURE);
#endif
#ifndef MCUBOOT_BUILTIN_KEY
    int key_id = -1;
#else
    /* Pass a key ID equal to the image index, the underlying
    crypto library
    * is responsible for mapping the image index to a builtin
    key ID.
    */
    int key_id = image_index;
#endif /* !MCUBOOT_BUILTIN_KEY */
#ifdef MCUBOOT_HW_KEY
    uint8_t key_buf[KEY_BUF_SIZE];
#endif
#endif /* EXPECTED_SIG_TLV */
    struct image_tlv_iter it;
    uint8_t buf[SIG_BUF_SIZE];
    uint8_t hash[IMAGE_HASH_SIZE];
    int rc = 0;
    FIH_DECLARE(fih_rc, FIH_FAILURE);
#ifdef MCUBOOT_HW_ROLLBACK_PROT
    fih_int security_cnt = fih_int_encode(INT_MAX);
    uint32_t img_security_cnt = 0;
    FIH_DECLARE(security_counter_valid, FIH_FAILURE);
#endif

    rc = bootutil_img_hash(enc_state, image_index, hdr, fap,
    tmp_buf,
        tmp_buf_sz, hash, seed, seed_len);
    if (rc) {
        goto out;
    }

    if (out_hash) {
        memcpy(out_hash, hash, IMAGE_HASH_SIZE);
    }

    rc = bootutil_tlv_iter_begin(&it, hdr, fap, IMAGE_TLV_ANY,
    false);
    if (rc) {
        goto out;
    }

```

```

    }

    if (it.tlv_end > bootutil_max_image_size(fap)) {
        rc = -1;
        goto out;
    }

    /*
     * Traverse through all of the TLVs, performing any
     checks we know
     * and are able to do.
     */
    while (true) {
        rc = bootutil_tlv_iter_next(&it, &off, &len, &type);
        if (rc < 0) {
            goto out;
        } else if (rc > 0) {
            break;
        }

        if (type == EXPECTED_HASH_TLV) {
            /* Verify the image hash. This must always be
            present. */
            if (len != sizeof(hash)) {
                rc = -1;
                goto out;
            }
            rc = LOAD_IMAGE_DATA(hdr, fap, off, buf,
            sizeof(hash));
            if (rc) {
                goto out;
            }

            FIH_CALL(boot_fih_memequal, fih_rc, hash, buf,
            sizeof(hash));
            if (FIH_NOT_EQ(fih_rc, FIH_SUCCESS)) {
                FIH_SET(fih_rc, FIH_FAILURE);
                goto out;
            }

            image_hash_valid = 1;
#ifdef EXPECTED_KEY_TLV

```

```

    } else if (type == EXPECTED_KEY_TLV) {
        /*
         * Determine which key we should be checking.
         */
        if (len > KEY_BUF_SIZE) {
            rc = -1;
            goto out;
        }
#ifdef MCUBOOT_HW_KEY
        rc = LOAD_IMAGE_DATA(hdr, fap, off, buf, len);
        if (rc) {
            goto out;
        }
        key_id = bootutil_find_key(buf, len);
#else
        rc = LOAD_IMAGE_DATA(hdr, fap, off, key_buf, len);
        if (rc) {
            goto out;
        }
        key_id = bootutil_find_key(image_index, key_buf, len);
#endif /* !MCUBOOT_HW_KEY */
        /*
         * The key may not be found, which is acceptable.
         There
         * can be multiple signatures, each preceded by a
         key.
         */
#ifdef EXPECTED_SIG_TLV
    } else if (type == EXPECTED_SIG_TLV) {
        /* Ignore this signature if it is out of bounds. */
        if (key_id < 0 || key_id >= bootutil_key_cnt) {
            key_id = -1;
            continue;
        }
        if (!EXPECTED_SIG_LEN(len) || len > sizeof(buf)) {
            rc = -1;
            goto out;
        }
        rc = LOAD_IMAGE_DATA(hdr, fap, off, buf, len);
        if (rc) {
            goto out;

```

```

    }
    FIH_CALL(bootutil_verify_sig, valid_signature, hash,
sizeof(hash),
                                buf, len, key_id);

    key_id = -1;
#endif /* EXPECTED_SIG_TLV */
#ifdef MCUBOOT_HW_ROLLBACK_PROT
    } else if (type == IMAGE_TLV_SEC_CNT) {
        /*
         * Verify the image's security counter.
         * This must always be present.
         */
        if (len != sizeof(img_security_cnt)) {
            /* Security counter is not valid. */
            rc = -1;
            goto out;
        }

        rc = LOAD_IMAGE_DATA(hdr, fap, off,
&img_security_cnt, len);
        if (rc) {
            goto out;
        }

        FIH_CALL(boot_nv_security_counter_get, fih_rc,
image_index,
                                &security_cnt);
        if (FIH_NOT_EQ(fih_rc, FIH_SUCCESS)) {
            FIH_SET(fih_rc, FIH_FAILURE);
            goto out;
        }

        /* Compare the new image's security counter value
        against the
         * stored security counter value.
         */
        fih_rc =
fih_ret_encode_zero_equality(img_security_cnt <
                                (uint32_t)fih_int_decode(security_cnt));
        if (FIH_NOT_EQ(fih_rc, FIH_SUCCESS)) {
            FIH_SET(fih_rc, FIH_FAILURE);
            goto out;
        }
    }
}

```

```

    }

    /* The image's security counter has been
    successfully verified. */
    security_counter_valid = fih_rc;
#endif /* MCUBOOT_HW_ROLLBACK_PROT */
}
}

rc = !image_hash_valid;
if (rc) {
    goto out;
}
#ifdef EXPECTED_SIG_TLV
    FIH_SET(fih_rc, valid_signature);
#endif
#ifdef MCUBOOT_HW_ROLLBACK_PROT
    if (FIH_NOT_EQ(security_counter_valid, FIH_SUCCESS)) {
        rc = -1;
        goto out;
    }
#endif

out:
    if (rc) {
        FIH_SET(fih_rc, FIH_FAILURE);
    }

    FIH_RET(fih_rc);
}

```

The provided C code contains several cryptographic operations and flash memory management functionalities, which are security-sensitive. Below is a **security review** of the code, focusing on potential vulnerabilities and best practices:

1. Flash Memory Handling Vulnerabilities

The code reads from flash memory (`flash_area_read`) and performs operations with data read from it. Potential vulnerabilities related to this are:

- **Unchecked Flash Reads:**

- If `flash_area_read` fails, the buffer content is not validated, and the program proceeds to operate on it. This could lead to undefined behavior if the buffer contains stale or uninitialized data.
- **Mitigation:** Add strict error checking and zero the buffer on failure (`memset` to 0). Example:

```
c

if (rc) {
    memset(tmp_buf, 0, blk_sz); // Avoid use of stale data
    bootutil_sha_drop(&sha_ctx);
    return rc;
}
```

- **Integer Overflow Risk in Offset Calculation:**

- Calculation of `off` and `blk_sz` in the loop could lead to integer overflows if `size` is large.
- **Mitigation:** Use safe integer operations and check for overflow conditions. Ensure that `blk_sz` does not exceed the buffer size.

2. Cryptographic Vulnerabilities

Use of SHA Context Without Finalizing in Error Paths

- In cases where `flash_area_read` fails, the SHA context is dropped using `bootutil_sha_drop()` without finalizing it via `bootutil_sha_finish()`.
- **Issue:** This could leave the SHA state in an undefined state, potentially causing subtle bugs in future calls.
- **Mitigation:** Ensure the `bootutil_sha_finish()` call is invoked before dropping the SHA context, even on failure paths.

Encryption Key Validation

- In the section:

c

```
if (MUST_DECRYPT(fap, image_index, hdr)) {  
    if (off >= hdr_size && off < tlv_off) {  
        blk_off = (off - hdr_size) & 0xf;  
        boot_encrypt(enc_state, image_index, fap, off - hdr_size,  
                    blk_sz, blk_off, tmp_buf);  
    }  
}
```

- **Potential Issue:** The `MUST_DECRYPT()` logic could be tampered with if an attacker influences image data to skip decryption logic. If the check fails, the code proceeds without decryption, leading to undefined behavior.
- **Mitigation:** Harden this condition using integrity checks to ensure the encrypted state is accurate and tamper-proof.

3. Signature Verification Weaknesses

Key Management Issues

- The function `bootutil_find_key()` compares the key hash to identify a valid key:

c

```
if (!memcmp(hash, keyhash, keyhash_len)) {  
    return i;  
}
```

- **Timing Attack Vulnerability:** `memcmp()` could introduce a timing side-channel vulnerability, leaking information about the key hash.
 - **Mitigation:** Use constant-time comparison functions like `boot_fih_memequal()` (already implemented elsewhere).

Multiple Signature Types

- If multiple signature types are allowed but improperly validated, this could lead to **signature confusion attacks**.

- **Mitigation:** Ensure that only one signature type is processed per image, as intended in the macro logic:

```
c

#if (defined(MCUBOOT_SIGN_RSA) + defined(MCUBOOT_SIGN_EC256) > 1)
#error "Only a single signature type is supported!"
#endif
```

Ensure this restriction remains enforced across builds.

4. Fault Injection Resistance Issues

- The use of fault injection mitigation (`FIH_CALL` , `FIH_DECLARE`) is good, but there is a risk that a critical path might still be tampered with.
- Example:

```
c

FIH_CALL(boot_fih_memequal, fih_rc, hash, buf, sizeof(hash));
if (FIH_NOT_EQ(fih_rc, FIH_SUCCESS)) {
    FIH_SET(fih_rc, FIH_FAILURE);
    goto out;
}
```

- **Potential Issue:** If a fault injection causes `boot_fih_memequal` to return incorrectly, the signature validation can be bypassed.
 - **Mitigation:** Ensure that all sensitive checks, especially those for signatures and keys, are hardened further with redundant fault-injection-resistant checks.

5. Inadequate Error Handling

- Several functions rely on returning `-1` or other non-zero values on failure. This could be ambiguous and lead to incorrect error propagation.

- **Mitigation:** Use well-defined error codes to differentiate between types of failures (e.g., `BOOT_EFLASH`, `BOOT_EBADARGS`).
-

6. Security Counter Validation Weaknesses

- In `bootutil_get_img_security_cnt()`:

```
c

if ((hdr == NULL) || (fap == NULL) || (img_security_cnt == NULL)) {
    return BOOT_EBADARGS;
}
```

- **Potential Issue:** If any of these parameters are NULL, this could indicate a memory corruption issue or a logic bug. Simply returning an error code may not be sufficient for safety.
 - **Mitigation:** Consider logging such events or halting the system for critical failures.
-

7. Recommendations for Secure Memory Handling

- **Zeroing Out Sensitive Data:** Ensure that cryptographic keys, hashes, and sensitive buffers are securely erased after use to prevent leakage through memory.
- Example:

```
c

memset(hash, 0, sizeof(hash)); // Zero out hash after use
```

Summary of Vulnerabilities and Mitigations

Issue	Risk	Mitigation
Unchecked flash reads	Undefined behavior	Validate buffers and zero on failure
Integer overflow in size calculations	Memory corruption, buffer overflows	Use safe arithmetic and bounds checking
Timing attack on key comparison	Side-channel attack	Use constant-time comparison functions
Fault injection risk	Bypass signature validation	Add redundant checks and hardening
Multiple signature types	Signature confusion attack	Enforce one signature type per image
Inadequate error handling	Ambiguous failures	Use clear and distinct error codes
Insufficient memory cleanup	Sensitive data leakage	Zero out buffers after use

This code contains multiple best practices (such as fault injection hardening), but careful attention is needed in error handling, cryptographic validation, and memory safety. Strengthening these areas will make the code more robust against potential attacks.

Provide the exact part of the code along with line numbers that are vulnerable or have security issues.

Here’s a detailed analysis with **line numbers** (approximated since the original code doesn't have line numbers) highlighting the potential security issues.

1. Unchecked Flash Reads

Location: Lines ~72–100

```
c

rc = flash_area_read(fap, off, tmp_buf, blk_sz);
if (rc) {
    bootutil_sha_drop(&sha_ctx);
}
```

```
    return rc;
}
```

- **Issue:** If `flash_area_read()` fails, the buffer `tmp_buf` remains in an undefined state, leading to potential stale data use.
- **Mitigation:** Zero out `tmp_buf` after a failed read:

```
c

if (rc) {
    memset(tmp_buf, 0, blk_sz); // Secure the buffer
    bootutil_sha_drop(&sha_ctx);
    return rc;
}
```

2. Integer Overflow in Size Calculations

Location: Lines ~63–70

```
c

for (off = 0; off < size; off += blk_sz) {
    blk_sz = size - off;
    if (blk_sz > tmp_buf_sz) {
        blk_sz = tmp_buf_sz;
    }
}
```

- **Issue:** If `size` is exceptionally large, `off + blk_sz` could overflow, leading to buffer overflows or incorrect reads.
- **Mitigation:** Add bounds checking and use safe integer operations.

3. Timing Attack on Key Comparison

Location: Lines ~341-348

```
c

if (!memcmp(hash, keyhash, keyhash_len)) {
    bootutil_sha_drop(&sha_ctx);
    return i;
}
```

- **Issue:** `memcmp()` is vulnerable to timing attacks.
- **Mitigation:** Use a constant-time comparison function:

```
c

FIH_CALL(boot_fih_memequal, fih_rc, hash, keyhash, keyhash_len);
```

yaml

```
---

### **4. Incomplete Fault Injection Hardening**
**Location: Lines ~558-566**
```c
FIH_CALL(boot_fih_memequal, fih_rc, hash, buf, sizeof(hash));
if (FIH_NOT_EQ(fih_rc, FIH_SUCCESS)) {
 FIH_SET(fih_rc, FIH_FAILURE);
 goto out;
}
```

- **Issue:** If a fault injection attack bypasses `boot_fih_memequal`, the program may accept an invalid signature.
- **Mitigation:** Add redundant checks to ensure signature integrity.

---

## 5. Multiple Signature Types Confusion

### Location: Lines ~282-293

c

```
#if (defined(MCUBOOT_SIGN_RSA) + defined(MCUBOOT_SIGN_EC256) > 1)
#error "Only a single signature type is supported!"
#endif
```

- **Issue:** Multiple signature types are blocked via preprocessor directives. However, if this restriction is bypassed at runtime, signature confusion could occur.
  - **Mitigation:** Perform a runtime check to ensure only one signature type is used per image.
- 

## 6. Security Counter Validation Issue

Location: Lines ~506–526

c

```
if (len != sizeof(img_security_cnt)) {
 rc = -1;
 goto out;
}
```

- **Issue:** If `img_security_cnt` is not properly validated, rollback attacks could occur.
  - **Mitigation:** Use secure comparisons and ensure rollback counters are strictly incremented.
- 

## 7. Inadequate Error Handling

Location: Lines ~496–500

c

```
if ((hdr == NULL) || (fap == NULL) || (img_security_cnt == NULL)) {
 return BOOT_EBADARGS;
}
```

```
}
```

- **Issue:** A simple error return might not be sufficient if the issue indicates a critical failure.
  - **Mitigation:** Log or halt execution for critical failures.
- 

## 8. Sensitive Data Not Cleared from Memory

Location: Multiple Places (e.g., Lines ~98, 332, 526)

- Example:

```
c

uint8_t hash[IMAGE_HASH_SIZE];
...
bootutil_sha_finish(&sha_ctx, hash);
```

- **Issue:** Cryptographic materials such as `hash` are not cleared after use.
- **Mitigation:**

```
c

memset(hash, 0, sizeof(hash)); // Clear sensitive data
```

---

## Summary Table of Vulnerable Sections

Line Range	Issue	Mitigation
~72-100	Unchecked flash read	Zero buffer on failure
~63-70	Integer overflow risk	Use safe arithmetic
~341-348	Timing attack on key comparison	Use constant-time comparison
~558-566	Incomplete fault injection hardening	Add redundant checks
~282-293	Signature confusion attack potential	Add runtime checks
~506-526	Security counter validation weakness	Ensure strict counter increments

Line Range	Issue	Mitigation
~496–500	Inadequate error handling	Log critical failures
Multiple	Sensitive data not cleared from memory	Zero buffers after use

Addressing these issues will significantly improve the security of the code.

Please analyze the code for common security vulnerabilities if present and map them to OWASP Top 10 Web Application Security Risk category/categories.

Here is an analysis of the provided C code for common security vulnerabilities, along with mapping each issue to the relevant **OWASP Top 10 Web Application Security Risks (2021)** categories.

## 1. Insufficient Error Handling and Validation

### Location:

- Lines ~496–500:

```
c

if ((hdr == NULL) || (fap == NULL) || (img_security_cnt == NULL)) {
 return BOOT_EBADARGS;
}
```

- **Issue:** The code returns a general error ( `BOOT_EBADARGS` ) without sufficient logging or handling. This could allow critical logic flaws to remain undetected, potentially leading to unexpected behavior or system failure.
- **OWASP Mapping:**
  - **A01:2021 – Broken Access Control:** Insufficient error handling could allow bypassing of validation mechanisms if sensitive components like security counters or headers are mishandled.

**Mitigation:** Use specific error codes for critical failures and ensure that errors related to cryptographic or secure operations are logged or trigger alerts.

---

## 2. Cryptographic Failures (Insecure Hash Comparison)

**Location:**

- Lines ~341–348:

```
c

if (!memcmp(hash, keyhash, keyhash_len)) {
 bootutil_sha_drop(&sha_ctx);
 return i;
}
```

- **Issue:** Using `memcmp()` for cryptographic comparisons can introduce **timing side-channel attacks**, leaking information about the comparison result to an attacker.
- **OWASP Mapping:**
  - **A02:2021 – Cryptographic Failures:** Timing side-channels are often categorized as cryptographic failures, especially in cases where keys or hashes are involved.

**Mitigation:** Use a **constant-time comparison function** like `boot_fih_memequal()` to prevent timing attacks.

---

## 3. Security Misconfiguration (Improper Signature Handling)

**Location:**

- Lines ~282–293:

```
c

#if (defined(MCUBOOT_SIGN_RSA) + defined(MCUBOOT_SIGN_EC256) > 1)
#error "Only a single signature type is supported!"
#endif
```

- **Issue:** The code enforces a **single signature type at compile time**, but this is not enforced at runtime. If misconfigurations occur during build-time or deployment, it could lead to **signature confusion attacks**.
- **OWASP Mapping:**
  - **A05:2021 – Security Misconfiguration:** Improper configuration handling (compile-time vs. runtime) can lead to security risks, especially for critical cryptographic operations.

**Mitigation:** Add runtime checks to ensure that only one signature type is used per image, even if multiple types are compiled.

---

## 4. Incomplete Fault Injection Mitigation

**Location:**

- Lines ~558–566:

```
c

FIH_CALL(boot_fih_memequal, fih_rc, hash, buf, sizeof(hash));
if (FIH_NOT_EQ(fih_rc, FIH_SUCCESS)) {
 FIH_SET(fih_rc, FIH_FAILURE);
 goto out;
}
```

- **Issue:** The use of fault injection hardening ( `FIH_CALL` ) is good practice, but **redundant checks** are missing, which could make the system susceptible to **fault injection attacks**. An attacker could tamper with the execution flow to accept invalid signatures.
- **OWASP Mapping:**
  - **A04:2021 – Insecure Design:** Lack of redundant controls for fault injection risks indicates an incomplete security design.

**Mitigation:** Use multiple redundant checks in security-critical operations, such as signature validation, to mitigate fault injection attacks.

---

## 5. Inadequate Memory Management (Sensitive Data Leakage)

Location:

- Lines ~98, 332, 526 (Multiple occurrences):

```
c

uint8_t hash[IMAGE_HASH_SIZE];
...
bootutil_sha_finish(&sha_ctx, hash);
```

- **Issue:** Sensitive data, such as cryptographic hashes or keys, is stored in memory but not cleared after use. This could lead to **data leakage** if an attacker gains access to memory.
- **OWASP Mapping:**
  - **A03:2021 – Insecure Data Handling:** Failure to securely handle sensitive data in memory increases the risk of unauthorized disclosure.

**Mitigation:** Ensure that all buffers containing sensitive data are **zeroed out** after use with `memset()` :

```
c

memset(hash, 0, sizeof(hash)); // Clear sensitive data
```

---

## 6. Lack of Bounds Checking (Integer Overflow and Buffer Overflow Risks)

Location:

- Lines ~63–70:

```
c

for (off = 0; off < size; off += blk_sz) {
 blk_sz = size - off;
 if (blk_sz > tmp_buf_sz) {
 blk_sz = tmp_buf_sz;
```

```
}
}
```

- **Issue:** If `size` is too large, it could result in **integer overflow** in the calculation of `off + blk_sz`. This might lead to **buffer overflows**, resulting in undefined behavior or memory corruption.
- **OWASP Mapping:**
  - **A06:2021 – Vulnerable and Outdated Components:** While this code may not directly correspond to an external component vulnerability, failure to handle buffer sizes correctly is a common vulnerability pattern.

**Mitigation:** Use safe arithmetic functions and validate `size` before performing arithmetic:

```
c

if (size > UINT32_MAX - off) {
 return -1; // Prevent overflow
}
```

---

## 7. Insufficient Security Counter Validation (Rollback Protection)

**Location:**

- Lines ~506–526:

```
c

if (len != sizeof(img_security_cnt)) {
 rc = -1;
 goto out;
}
```

- **Issue:** If the security counter is not properly validated, it could allow **rollback attacks** (downgrading to a vulnerable firmware version).
- **OWASP Mapping:**
  - **A01:2021 – Broken Access Control:** Insufficient validation of security counters can allow unauthorized access to older, vulnerable firmware.

**Mitigation:** Implement stricter rollback protection by enforcing **monotonic counter increments** and performing secure comparisons.

---

## 8. Insufficient Handling of Flash Read Errors

**Location:**

- Lines ~72–100:

```
c

rc = flash_area_read(fap, off, tmp_buf, blk_sz);
if (rc) {
 bootutil_sha_drop(&sha_ctx);
 return rc;
}
```

- **Issue:** If the flash read fails, the buffer content remains in an undefined state. This can lead to the use of **stale or uninitialized data**, which could compromise integrity.
- **OWASP Mapping:**
  - **A08:2021 – Software and Data Integrity Failures:** The use of unvalidated or stale data increases the risk of software integrity issues.

**Mitigation:** Zero out the buffer if the read fails to prevent undefined behavior:

```
c

if (rc) {
 memset(tmp_buf, 0, blk_sz); // Clear buffer on failure
 bootutil_sha_drop(&sha_ctx);
 return rc;
}
```

---

## Summary Table with OWASP Mapping

Issue	Line Range	OWASP Top 10 Mapping	Description
Insufficient error handling	~496–500	A01:2021 – Broken Access Control	Improper error handling could allow bypassing validations.
Timing attack on key comparison	~341–348	A02:2021 – Cryptographic Failures	<code>memcmp()</code> used, leaking information through timing.
Signature confusion attack potential	~282–293	A05:2021 – Security Misconfiguration	Only compile-time checks used for signature types.
Incomplete fault injection mitigation	~558–566	A04:2021 – Insecure Design	Fault injection hardening is insufficient.
Sensitive data not cleared from memory	~98, 332, 526	A03:2021 – Insecure Data Handling	Sensitive data not properly zeroed out after use.
Integer overflow and buffer overflow risk	~63–70	A06:2021 – Vulnerable and Outdated Components	Arithmetic errors can lead to buffer overflows.
Insufficient rollback protection	~506–526	A01:2021 – Broken Access Control	Weak security counter validation allows rollback attacks.
Unchecked flash read errors	~72–100	A08:2021 – Software and Data Integrity Failures	Stale data can compromise software integrity.

## Conclusion

This code exhibits several common security vulnerabilities that align with the **OWASP Top 10 Web Application Security Risks**. Addressing these issues—such as improving cryptographic operations, ensuring proper error handling, and implementing stronger memory management—will significantly enhance the security posture of the code.

For each identified vulnerability, provide an exploit example that an attacker could use. Include a step-by-step explanation of how the exploit works, the necessary conditions or setup, any potential limitations or obstacles an attacker might face, and the likelihood of this attack happening. Ensure the exploits are specific to the vulnerability type provided.

Here are specific exploit scenarios for each identified vulnerability. These include step-by-step explanations, conditions required, potential obstacles, and likelihood assessments.

---

## 1. Insufficient Error Handling

### Exploit Scenario:

- **Vulnerability:** Improper error handling allows critical security logic to be bypassed.
- **Exploit Example:** An attacker exploits missing error logs to repeatedly inject malformed `hdr` structures into `bootutil_get_img_security_cnt`.

### Steps:

1. The attacker provides a corrupt or malformed `hdr` parameter in the firmware image header.
2. The function `bootutil_get_img_security_cnt()` fails and returns `BOOT_EBADARGS`.
3. Since no alert or critical failure occurs, the attacker uses this repeatedly to identify weak points in firmware verification logic.

### Conditions:

- The system relies on silent error handling without logging or alert mechanisms.

### Limitations:

- Requires access to image input handling or firmware injection mechanisms.
- Exploit is indirect and depends on subsequent failures to materialize issues.

### Likelihood:

**Low.** Exploiting error handling gaps often requires significant context knowledge and insider information.

---

## 2. Cryptographic Failures (Timing Attack on Key Comparison)

### Exploit Scenario:

- **Vulnerability:** `memcmp()` used for cryptographic key comparisons is vulnerable to timing attacks.
- **Exploit Example:** An attacker measures the time taken by the firmware to reject invalid key hashes during the boot process.

#### Steps:

1. The attacker repeatedly sends crafted key hashes to the firmware loader.
2. By analyzing timing differences, the attacker determines how much of the key hash matches the expected value.
3. Over time, the attacker reconstructs the expected hash.

#### Conditions:

- The attacker must have physical access to observe precise timing or use a side-channel (e.g., power analysis) to detect timing differences.

#### Limitations:

- Requires high-resolution timing measurements.
- May need physical proximity to the device.

#### Likelihood:

**Moderate.** Timing attacks are well-documented but require expertise and resources.

---

### 3. Security Misconfiguration (Signature Confusion)

#### Exploit Scenario:

- **Vulnerability:** Runtime verification does not enforce a single signature type.
- **Exploit Example:** The attacker injects an image signed with a less secure algorithm (e.g., RSA-1024 if it exists).

#### Steps:

1. The attacker crafts a firmware image signed with a deprecated or less secure signature type.
2. The attacker introduces this image into the device's update process.

3. Without runtime enforcement, the system may incorrectly accept the weaker signature.

**Conditions:**

- Requires the ability to craft or introduce malicious firmware images.

**Limitations:**

- Dependent on system misconfiguration and lack of runtime signature type enforcement.

**Likelihood:**

**Moderate to High.** Misconfigurations are a common exploit vector.

---

## 4. Incomplete Fault Injection Mitigation

**Exploit Scenario:**

- **Vulnerability:** Insufficient hardening against fault injection during critical operations like signature validation.
- **Exploit Example:** The attacker injects faults to skip or bypass signature validation.

**Steps:**

1. The attacker uses a physical method (e.g., power glitches or electromagnetic pulses) to induce faults during `bootutil_img_validate()`.
2. The fault causes the system to skip over critical checks (e.g., `FIH_CALL(boot_fih_memequal)`).
3. The device boots an unsigned or maliciously signed firmware image.

**Conditions:**

- Physical access to the device and knowledge of when to inject the fault.

**Limitations:**

- Requires specialized equipment and precise timing.

**Likelihood:**

**Low to Moderate.** While fault injection is powerful, it's resource-intensive and requires expertise.

---

## 5. Inadequate Memory Management (Sensitive Data Leakage)

### Exploit Scenario:

- **Vulnerability:** Sensitive data, such as cryptographic hashes, is not cleared from memory after use.
- **Exploit Example:** An attacker uses memory scraping to extract cryptographic hashes.

### Steps:

1. The attacker compromises the system via a secondary exploit (e.g., code injection) or physical access.
2. They dump memory regions containing sensitive data (e.g., `hash[IMAGE_HASH_SIZE]` ).
3. Using the leaked hash, the attacker crafts a malicious firmware image with matching hash values.

### Conditions:

- Requires memory access or the ability to run arbitrary code.

### Limitations:

- Dependent on an additional vulnerability to access memory.

### Likelihood:

**Moderate.** Sensitive data leakage is common in memory-constrained systems without secure wiping mechanisms.

---

## 6. Integer Overflow and Buffer Overflow Risks

### Exploit Scenario:

- **Vulnerability:** Integer overflow in the loop handling offsets ( `off` ) and buffer sizes ( `blk_sz` ).
- **Exploit Example:** The attacker provides a specially crafted `size` value to trigger a buffer overflow.

### Steps:

1. The attacker crafts an image with a `size` value close to the maximum value of a 32-bit integer.
2. During the loop in `bootutil_img_hash`, `off + blk_sz` overflows, resulting in out-of-bounds memory access.
3. The attacker injects payloads into adjacent memory regions to overwrite security-critical data.

### Conditions:

- The attacker must control `hdr->ih_img_size` or a similar size-related variable.

### Limitations:

- Requires precise control over crafted image data.

### Likelihood:

**Moderate to High.** Integer overflows are a common exploit target.

---

## 7. Insufficient Security Counter Validation

### Exploit Scenario:

- **Vulnerability:** Weak validation of rollback counters.
- **Exploit Example:** An attacker injects an older firmware image with a lower security counter.

### Steps:

1. The attacker identifies a device running a newer firmware version.
2. They inject an older firmware image with a valid signature but a lower security counter.
3. The device accepts the rollback image due to incomplete counter validation.

### Conditions:

- The attacker must be able to inject firmware into the update process.

### Limitations:

- Dependent on the absence of additional rollback protection measures.

**Likelihood:**

**High.** Rollback attacks are a well-known and widely exploited issue in embedded systems.

---

**8. Unchecked Flash Read Errors**

**Exploit Scenario:**

- **Vulnerability:** Failure to zero buffers on flash read errors.
- **Exploit Example:** The attacker manipulates the flash area to inject stale or invalid data.

**Steps:**

1. The attacker tampers with the flash memory to corrupt part of the firmware image.
2. When `flash_area_read()` fails, `tmp_buf` contains stale or uninitialized data.
3. The attacker ensures that the stale data aligns with a malicious payload to exploit downstream processing logic.

**Conditions:**

- Requires the ability to tamper with flash storage.

**Limitations:**

- Exploit depends on specific implementation details downstream.

**Likelihood:**

**Moderate.** Flash manipulation attacks are feasible but require some technical sophistication.

---

**Summary Table**

Vulnerability	Exploit Example	Likelihood	Requirements
Insufficient Error Handling	Malformed <code>hdr</code> causes silent bypass	Low	Access to input-handling logic

Vulnerability	Exploit Example	Likelihood	Requirements
Cryptographic Failures (Timing Attack)	Measure timing to reconstruct key hash	Moderate	Physical access and side-channel
Security Misconfiguration	Inject weaker signature types	Moderate to High	Ability to craft malicious firmware
Fault Injection Risks	Bypass signature validation with faults	Low to Moderate	Specialized equipment
Inadequate Memory Management	Extract cryptographic hash from memory	Moderate	Memory scraping or dumping
Integer and Buffer Overflows	Out-of-bounds memory access through crafted size	Moderate to High	Control over size variables
Insufficient Rollback Protection	Downgrade firmware using older image	High	Firmware injection
Unchecked Flash Read Errors	Stale data injected into uninitialized buffer	Moderate	Flash tampering capabilities

By addressing these vulnerabilities, the attack surface of the system can be significantly reduced, enhancing overall security.

Is this conversation helpful so far?